T-Mobile Fit4Launch WiFi Calling (WFC)
Lab Test Plan

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Table of Contents

1. Revision History

Revision	Date	Purpose	Editor
1.0	Feb 2021	Initial Version of VoWiFi Handover Test plan for 2021 including 5G	Brett Day
1.1	April 2021	Included T-Mobile feedback	Brett Day
1.5	April 2021	Reviewed with PQE team	Brett Day
1.6	April 2021	Official first draft of the test plan	Brett Day
1.7	September 2021	Added comments for change	Brett Day
1.8	January 2022	Added Multi Handover Profiles and modified number of APs from 3 to 4	Adnan Sana / Brett Day
2.0	May 2022	Added Pass/Fail Criteria	Adnan Sana / Brett Day
2.1	August 2023	Added Reference Device Added Baseline Audio Performance Updated Pass / Fail Criteria Added Baseline Cellular and WFC Call Performance Updated Impaired WiFi Call Performance (WiFi Only) Added T-Mobile Nokia FWA Gateway WFC Performance Updated Multi Handover Test Description Updated Minimum Device Requirements for Testing	Adnan Sana / Brett Day

Table 1 - Revision History

2. Introduction

The principal objective of the WiFi Calling (WFC) or Voice Over WiFi (VoWiFi) testing executed by Spirent is to objectively compare the quality of experience performance of a subject device in a live WiFi/Cellular network and/or laboratory environment using Umetrix and octoScope products.

Empirical data captured during the test is aimed at measuring Call Performance, Handover Success, and Audio Quality in terms of POLQA MOS. The MOS metrics are collected with natural impairments over the live network and with simulated impairments to evaluated primary questions:

- Does the Device Under Test (DUT) operate properly in various RF conditions, on WiFi and/or the cellular network or when exposed to a simulated impairments network environment compared to a Reference Device?
- Does the DUT perform comparable to the reference device under all conditions with equal or better Audio MOS Quality performance while being used in the same way as a user would use it?

Note: All Lab testing will be conducted at Spirent Performance Lab located 5280 Corporate Drive, Suite A100, Frederick, MD 21703.

2.1. Reference Device

The Reference device for WFC Testing is Samsung GS23.

3. Test Plan Overview

To deliver key performance indicators for the device under study, the following testing will be conducted:

Note: Device on the Cellular Preferred setting is required for all tests.

- **Functional Verification** – Spirent performs Functional tests on all devices that are received for testing to include audio capabilities with test equipment, voice calling over 4G/5G network, and voice calling over WiFi access points used for testing.

- **WFC Test** – Spirent provides an access point for four (4) common marketplace routers. The goal of this test is to identify differences in performance with and without IP impairments.

  Here is the list of 4 Access Points to be used:

  - 5G21-12W-A (Nokia) – WiFi 6
  - ASUS RT-AC68U
  - Google Nest Wi-Fi AP AC2200
  - Linksys Hydra Pro 6E (Linksys Hydra Pro 6E (AXE6600)) – WiFi 6E

  4G/5G Cellular Network Test – Spirent utilizes common workplace environment to establish voice calling over live network on 4G/5G signal. The goal of this test is to identify performance of voice over 4G/5G calling, including VoNR/VoLTE to WiFi.

- **Handover Test** – Spirent utilizes a common work environment along with four (4) common marketplace routers as access points (APs) to establish voice calling on the device under test. Devices are required to handover from Live network to WiFi AP and from WiFi AP to 4G/5G network. Handovers will be triggered in two ways, 1) RF Signal attenuation and 2) IP impairments introduced in a controlled manner utilizing a Spirent Attero network emulator in line with the access point under test. The DUT will be tested in both 5G and 4G network coverage.

- **Summary of Reporting** – Spirent will produce a report for each scenario listed above. Each section of a report shall have key performance indicators (KPIs) which is to serve as overall device performance. From these KPIs performance will be understood.

The following sections describe the Full Test components.

3.1. Full Test Components Overview

3.1.1. Functional Verification

The purpose of functional verification is to make sure devices are in working condition and can be tested.

- Voice calls over WiFi and Cellular for a standard setup.
- Establish that WFC satisfies min requirements under ideal conditions and that the device is capable of being tested.

3.1.2. Baseline Audio Performance

The purpose of Baseline Audio Performance is to objectively measure the Active Speech level and frequency response of a Cellular and WiFi voice call to ensure devices are properly aligned between the two radio access technologies.

- Active Speech level is measured by using ITU-T p.56 algorithm.
- An audio sample is collected from a Voice over WiFi and a cellular call.
- Active speech level is measured for both calls and delta calculated.
- Each audio sample's frequency response is compared to each other and source reference file.
- The following KPIs are reported as well: p.56 active speech level, p.56 rms long term energy, p.56 active peak-factor and POLQA Attenuation.
- The pass and fail criteria are based on the p.56 active speech level delta between cellular and WiFi call audio sample defined in Section 5 of this document.

3.1.3. Baseline Cellular Call Performance

The purpose of Voice calling is to ensure that the device under test can maintain call connectivity and Speech Quality (MOS) over the live network.

- The device will be tested using Live network signal. The device will be placed in auto mode for cellular signal.
- For Cellular Baseline Call Performance five mobile to mobile calls are performed on both REF and DUT devices.
- Device placed in a small chamber.
  - RF piped in from cellular 4G/5G network
- Spirent's Umetrix Voice is used to conduct this test .
- KPIs measured: Mean Setup time(s), Successful Initiations (%), Dropped Calls (%), MOS (MO) and MOS (MT)
- There is no pass / fail criteria for Baseline Cellular Call Performance.

3.1.4. Baseline WFC Call Performance (No Impairments)

The purpose of Baseline WFC Call Performance is to ensure that a DUT is able to make calls over WiFi network without IP impairments.

- For WFC Baseline Call Performance five mobile to mobile calls are performed on both REF and DUT devices using LinkSys AP.
  - Call Duration 90 sec with 30 secs wait time
- Spirent's Umetrix Voice is used to conduct this test.
  - In total 45 MOS samples are collected and then averaged for reporting for both MO and MT DUT and REF devices
- KPIs measured: Mean Setup time(s), Successful Initiations (%), Dropped Calls (%), MOS (MO) and MOS (MT).
- Pass / Fail criteria are applied on the Baseline WFC performance based on criteria defined in Section 5.

3.1.5. Impaired WiFi Call Performance (WiFi Only)

The purpose of this testing is to document DUT’s WiFi Call Performance statistics (AFR/DCR) and measure Audio Quality during IP impairments across a range of popular APs (compatibility).

- Device will be tested using three common workplace APs:
  - ASUS RT-AC68U
  - Google Nest Wi-Fi AP AC2200
  - Linksys Hydra Pro 6E
- Each AP shall have IP impairments introducing packet loss, jitter and gaussian distribution as per the following 3 Profiles:
  - ETSI-B (Standard TR 126)
  - NSD-A (T-Mobile defined)
  - NSD-C (T-Mobile defined)

ETSI-B	- Gaussian distribution:	Packet Loss: 3%	Downlink Jitter: σ = 40 ms

Table 2 ETSI Impairments

NSD-A	- Gaussian distribution:	Packet Loss: 8%	Downlink Jitter: σ = 10 ms
NSD-C	- Gaussian distribution	Packet Loss: 15%	Downlink Jitter: σ = 60 ms

Table 3 Non-Standard Impairments

- Each AP shall have WiFi signal at a common range a user would experience in office or home setting.
- WFC Call Performance testing will be performed on 5GHz WiFi spectrum. In case a device does not support 5GHz, 2.4GHz will be substituted.
- The RSSI shall be -65 dBm at the beginning of each test scenario.
  - Device placement includes use of turntable (continuous rotation at 0.25 rpm)
  - Device placed in a small chamber to isolate from environment variables
  - Please note a signal fluctuation of +/- 3 dB in RSSI is commonly expected due to AP/device antenna RF coupling

WiFi Call Performance (WiFi Only) with 3 Impairment Profiles							
APs	ETSI-B	NSD-A	NSD-C	Total Calls	Call Duration/Call	Wait Time/Call	MOS Samples
ASUS RT-AC68U (DUT & REF)	24	24	24	72	90 secs	30 secs	648 (216/profile)
Google Nest AC2200 (DUT & REF)	24	24	24	72	90 secs	30 secs	648 (216/profile)
Linksys Hydra Pro 6E (DUT & REF)	24	24	24	72	90 secs	30 secs	648 (216/profile)

Table 4 Impaired WiFi Call Performance

- KPIs measured: Mean Setup time(s), Successful Initiations (%), Dropped Calls (%), MOS (MO) and MOS (MT).
- Pass / Fail criteria is applied on the KPIs as per Section 5 of this document.

3.1.5.1. T-Mobile Nokia 5G FWA Gateway WFC Performance Testing (No Impairments)

The purpose of this test case is to ensure that a DUT is compatible with T-Mobile's FWA Nokia 5G Gateway without simulated impairments. FWA Gateway's WiFi service is backboned to T-Mobile's 5G/4G Cellular live network.

AP	Impairment Profile	Total Calls	Call Duration/Call	Wait Time/Call	MOS Samples
5G21-12W-A (Nokia FWA) (DUT & REF)	No Impairment	24	90 secs	30 secs	216

Table 5 Nokia 5G FWA Gateway WiFi Call Performance

- KPIs measured: Mean Setup time(s), Successful Initiations (%), Dropped Calls (%), MOS (MO) and MOS (MT).
- Pass / Fail criteria is applied on the KPIs as per Section 5 of this document.

3.1.6. Handover Tests

The purpose of Handover Tests is to ensure that a DUT hand overs between WFC to Cellular and vice versa. The objective is to make sure a user has uninterrupted service. The handover tests are performed using combination following four APs:

- 5G21-12W-A (Nokia) – WiFi 6
- ASUS RT-AC68U
- Google Nest Wi-Fi AP AC2200
- Linksys Hydra Pro 6E (Linksys Hydra Pro 6E (AXE6600)) – WiFi 6E

All the handover test cases are performed on 2.4GHz WiFi spectrum.

Each test case for all the Handover tests will be allowed for a maximum of 2 attempts to pass a test case. If a device fails to perform a handover test case 2 times, a note will be mentioned in the report stating the fact.

The handover thresholds criteria defined by T-Mobile is as follows:

- Cellular to WiFi Handover
    - WiFi RSSI >= -70 dBm
    - Cellular RSRP <= -114 dBm
- WiFi to Cellular Handover
    - Cellular RSRP >= -110 dBm
    - WiFi RSSI <= -80 dBm

3.1.6.1. Multi Handover (Profile 1, 2 & 3)

To ensure that the device can handover multiple times in a single call at defined thresholds.

- Three Handover Profiles (1, 2 and 3) based on RF signal degradation on Cellular and WiFi on 3 APs:
    - 5G21-12W-A (Nokia) – WiFi 6
    - ASUS RT-AC68U
    - Linksys Hydra Pro 6E (Linksys Hydra Pro 6E (AXE6600)) – WiFi 6E
- Cellular RF Ranges vary between RSRP -90 to -120 dBm at different intervals.
- WiFi RF Ranges vary between RSSI -40 dBm to no WiFi signal present at different intervals.
- Profile 1 & 2 shall start the test with DUT connected via WFC.
- Profile 3 shall start the test with the DUT connected via Cellular.
- Single call test.
- Device will be stressed for 20 mins under variable signal conditions per profile and per AP
- KPIs measured: Successful Handover Count, Call Drops and Speech Quality (MOS)
- Pass / Fail criteria is applied on the KPIs as per Section 5 of this document.

3.1.6.2. IP Impairments (Profile 4)

To measure the effects of IP impairments on WFC to Cellular handovers on following 3 APs:

- ASUS RT-AC68U
- Google Nest Wi-Fi AP AC2200
- Linksys Hydra Pro 6E (Linksys Hydra Pro 6E (AXE6600)) – WiFi 6E

- For each AP, IP impairments will be introduced to force device to handover to Cellular network.
    - Each AP will be broadcast a good WiFi signal to DUT (RSSI ranges between -60 to -70 dBm) at the beginning of the test
    - Packet loss and jitter shall be introduced with values increasing every 20 seconds
        - Every 20 seconds
            - Packet loss in downlink increase by 1%
            - Jitter +2 ms
- Samples shall be collected in a mobile-to-mobile scenario.

- 10 MOS samples are collected before handover and up to 10 MOS samples during/after handover.
- Handover latency shall be measured using white noise continuous audio file of 48kHz sample rate.
  - Latency silence shall be measured and reported based on this test file
  - KPI Measure duration (ms) of any loss of audio within the audio file at time of handover
  - Signal strength when handover occurred shall be reported
- Cellular signal shall be in range between -110 to -115 dBm.
- KPIs measured: Speech Quality (MOS) before Handover and MOS During/After Handover, Packet Loss at Handover (%), Jitter at Handover and Handover Speech Delay (audible silence) at Handover (sec).
- Pass /Fail criteria is applied on the KPIs as per Section 5 of this document.

3.1.6.3. Walk out of the WiFi Coverage (Profile 5)

To ensure handovers occur when the serving WiFi signal diminishes and handover to Live cellular network (LTE/5G)

- This is an RF impairment scenario.
- WFC to Cellular Network handover.
- Profile 5 is performed on following 3 APs:
  - 5G21-12W-A (Nokia) – WiFi 6
  - ASUS RT-AC68U
  - Linksys Hydra Pro 6E (Linksys Hydra Pro 6E (AXE6600)) – WiFi 6E

- For each AP, RF impairment will be introduced to force the device to handover to the Cellular network.
  - Each AP will start at a WiFi signal at RSSI -65 dBm. From this signal strength, signal shall be attenuated
  - Profile 5: Attenuation steps / duration (Start at -65 dBm, increase attenuation every 20 seconds until WiFi signal is not present simulating a user walking out of WiFi Coverage.
- Samples shall be collected in a mobile-to-mobile scenario.
- Single call test.
- 10 MOS samples are collected before handover and up to 10 MOS samples during/after handover.
- Handover latency shall be measured using white noise continuous audio file of 48kHz sample rate.
  - Latency silence shall be measured and reported based on this test file
  - KPI Measure duration (ms) of any loss of audio within the audio file at time of handover
  - Signal strength when handover occurred shall be reported
  - Cellular signal shall be in a range between -110 and -115 dBm
- KPIs measured: Speech Quality (MOS) before Handover and MOS During/After Handover, and Handover Speech Delay (audible silence) at Handover (sec), WiFi RSSI (dB), Call Drops and Handover Attempts of Max of 2 allowed.
- Pass / Fail criteria is applied on the KPIs as per Section 5 of this document

3.1.6.4. Walk into WiFi Coverage (Profile 6)

To ensure handovers take place when user moves from Live Cellular (LTE/5G) into a stronger WiFi signal.

- This is an RF impairment scenario.
- Cellular network calling to WFC handover.
- Profile 5 is performed on following 3 APs:
  - 5G21-12W-A (Nokia) – WiFi 6
  - ASUS RT-AC68U
  - Linksys Hydra Pro 6E (Linksys Hydra Pro 6E (AXE6600)) – WiFi 6E

- For each AP, RF impairment will be introduced to force device to handover to WiFi AP.
  - Each test will start with device on a cellular call
  - Profile 6: WiFi signal will be less than -85 dBm RSSI to start the test. WiFi signal will increase every 20 secs until the end of the test to replicate a consumer walking into stronger WiFi coverage
  - Cellular Attenuation steps / duration starting with good RSRP greater than -105 dBm and as WiFi signal strength increases Cellular signal will be attenuated at different intervals.

- Samples shall be collected in a mobile-to-mobile scenario.
- Single call test.
- 10 MOS samples are collected before handover and up to 10 MOS samples during/after handover.
- Handover latency shall be measured using white noise continuous audio file of 48kHz sample rate.
    - Latency silence shall be measured and reported based on this test file
    - KPI Measure duration (ms) of any loss of audio within the audio file at time of handover
    - Signal strength when handover occurred shall be reported
    - Cellular signal shall be in a range between -110 and -115 dBm
    - WiFi signal will gradually increase in signal strength
- KPIs measured: Speech Quality (MOS) before Handover and MOS During/After Handover, and Handover Speech Delay (audible silence) at Handover (sec), WiFi RSSI (dB), Call Drops and Handover Attempts of Max of 2 allowed.
- Pass / Fail criteria is applied on the KPIs as per Section 5 of this document.

3.2. Testing Guidelines

Please adhere to the following guidelines regarding scheduling and hardware requirements to ensure successful completion of the tests.

3.2.1. Scheduling

Please notify Spirent at least five (5) business days before we are to receive shipment of a new device.

3.2.2. Hardware Requirements

The minimum quantity of devices required, and test completion SLA is listed in the table below.

	SLA Duration in business days	Minimum Quantity
WFC and Live Cellular Network	15 days	6 Normal form factor devices

Table 6 Device quantity based on SLA

Spirent offers an expedited test completion delivery within 10 business days with a 30% upcharge fee on the standard List Price.

Spirent also requires the following items to complete the tests efficiently:

- USB data cables and chargers
- Steps to set the network to LTE, 5G NR on the device as required for the test
- Visible network indicators to identify the network serving the device
- Device drivers

Hardware and Software Versions: OEMs must provide steps to verify the hardware and software versions on the device. All devices submitted for F4L and WFC testing must have the same hardware version and software version. Any deviation from this requirement will need a written approval from T-Mobile.

4. Test Summary

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Test Description	Variables	Desired Outcome
Functional Verification	Voice calls over Wi-Fi and Cellular for a standard setup	Establish that WFC satisfies min requirements under ideal conditions and that the device is capable of being tested
Baseline Cellular and WiFi Calling Tests	Device will be tested using WiFi and Live Cellular network signal	To ensure that device can maintain call connectivity and Speech Quality (MOS) over the WiFi and Live Cellular network
Impaired WFC Test (WiFi only)	Three Access Points (APs), three IP Impairment profiles (Std B, Non Std A, C)	Document devices WFC AFR/DCR during IP impairments and measure Audio Quality and Call Performance across a range of popular APs (compatibility)
Handover Tests (WFC→ Cellular/Cellular to WFC)	RF Signal range, Packet Loss (**6** profiles)	Measure Speech Latency, Audio quality and AFR/DCR before, during and after HO in both directions

Table 5 – WFC Test Plan Summary

5. Pass and Fail Criteria

Baseline Audio Performance	
p.56 Audio Active Speech Level (ASL) Delta	ASL <= 2
	ASL > 2 and < 3.5
	ASL >= 3.5

Call Performance WFC Baseline, ETSI-A, NSD-B, NSD-C	
Mean Setup Time	if(Avg of DUT) < (Avg of REF)
	if(Avg of DUT) < (Avg of REF*1.1)
	if(Avg of DUT) between (Avg of REF)*1.25 & 1.1
	if(Avg of DUT) > (Avg of REF*1.25
Call Success/Drops	If p-value > 0.95 for DUT vs REF
	If p-value between 0.05 and 0.95 for DUT vs REF
	RED condition is satisfied but % failures/Drops of DUT < 1%
	If p-value for Fisher Exact < 0.05 for DUT vs REF
MOS	If DUT > REF
	If DUT > REF-0.1
	If Red or Green Not Satisfied
	If DUT<REF-0.25

RF Multi Handover	
MOS	If DUT > REF
	If DUT > REF-0.1
	If Red or Green Not Satisfied
	If DUT<REF-0.25
WFC RSSI	If WFC RSSI >= -73
	If WFC RSSI < -73
Cellular RSSP	If Cellular RSRP >= -119
	If Cellular RSRP < -119
Minimum Handovers	If No of Handovers >=5
	If No of Handovers < 5
Call Drops	If Call Drops = 0
	If Call Drops = 1
	If Call Drops = 2

RF Handover WFC to Cellular	
MOS	If DUT > REF
	If DUT > REF-0.1
	If Red or Green Not Satisfied
	If DUT<REF-0.25
Handover Impact	If Avg of Seconds <= 2
	If Avg of Seconds > 2
RSSI	If RSSI <= -77
	If RSSI > -77
Call Drops	If Call Drops = 0
	If Call Drops = 1
	If Call Drops = 2

RF Handover Cellular to WFC	
MOS	If DUT > REF
	If DUT > REF-0.1
	If Red or Green Not Satisfied
	If DUT<REF-0.25
Handover Impact	If Avg of Seconds <= 2
	If Avg of Seconds > 2
RSSI	If RSSI >= -73
	If RSSI < -73
Call Drops	If Call Drops = 0
	If Call Drops = 1
	If Call Drops = 2

IP Impairment Handover	
Packet Loss	If % Packet Loss <= 35%
	If % Packet Loss > 35%
MOS	If DUT > REF
	If DUT > REF-0.1
	If Red or Green Not Satisfied
	If DUT<REF-0.25
Call Drops	If Call Drops = 0
	If Call Drops = 1
	If Call Drops = 2
Handover Speech Delay (audible silence)	If < 2 secs
	If >= 2 sec